

Arnab Bhattacharjee

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Research Interests

Human-Computer Interaction (HCI), Agentic AI, Computer Science Education (CSEd), Human-AI Interaction, Systems

Education

Bangladesh University of Engineering and Technology (BUET)

February 2020 - March 2025

B.Sc. in Computer Science and Engineering

CGPA: 3.88 / 4.00

- Thesis: *Evaluating the Multilingual Capabilities of LLM-based Web UI Agent*, supervised by Dr. Rifat Shahriyar
- Dean's List Scholarship (Level 1 and Level 2)

Notre Dame College, Dhaka

June 2017 - May 2019

Higher Secondary Certificate (HSC)

GPA: 5.00 / 5.00

Research Experience

Human-AI Co-Creation for Technical Lecture Content | *HCI, Agentic AI, CS Education*

2026

- Investigated how computer science educators author technical lecture content with generative AI tools.
- In a formative study, conducted semi-structured interviews with university educators and analyzed them using reflexive thematic analysis to identify their challenges and derive design goals.
- Guided by these goals, built an agentic pipeline that generates technical lecture content under the educator's direction.
- Conducted a qualitative user evaluation through focus group discussions (FGDs) with educators.

Evaluating the Multilingual Capabilities of LLM-based Web UI Agent | *Undergraduate Thesis, BUET*

2024 - 2025

- Developed a multilingual web agent that navigates web pages and performs actions to complete complex, multi-step tasks given by a user.
- Fine-tuned a multilingual model (mT5-base) on an extended Mind2Web dataset, and built web agents on three frontier LLMs (Gemini-2.0-Flash, GPT-4o-mini, DeepSeek-V3) to evaluate their zero-shot performance.
- Built a VLM-based automated evaluation system for web agents.
- Supervised by Dr. Rifat Shahriyar, Professor, BUET.

Language Effects on LLM Web Agents | *LLM Agents, Multilingual Evaluation*

- Evaluated an HTML-based LLM web agent across several widely spoken languages in two settings: a controlled offline benchmark in which the page structure stays fixed and only the language changes, and live task execution on real localized websites.
- Quantified how much of the agent's observation actually changes with the language, alongside task performance.
- Found that performance stayed close to English even when most of the page text was rewritten, while non-English pages consumed a substantially larger share of the model's context budget.

Poster

Abir Muhtasim*, **Arnab Bhattacharjee***, Abhik Bhattacharjee, Faria Huq, Rifat Shahriyar. *How well do LLM Web Agents work for non-English Language?* LLM Agent Workshop, Carnegie Mellon University, Pittsburgh, PA, USA. May 2, 2024. Poster, Extended Abstract. (*Equal contribution)

Work Experience

United International University (UIU)

July 2025 - Present

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

- Courses taught: Data Structures and Algorithms, Discrete Mathematics, Object-Oriented Programming, Operating Systems, Digital Logic Design, Network Security, and Social, Environment and Engineering Ethics.
- Course coordinator for Microprocessor and Microcontroller (theory and lab), including embedded systems projects with Arduino and Raspberry Pi.

Samsung Research and Development Institute, Bangladesh

May 2025 - July 2025

Software Engineer

Dhaka, Bangladesh

- Developed an Android application (Kotlin, Android Studio) for an intelligent pen that writes on any surface, with the strokes appearing on the Android device automatically.

Projects

Machine Learning Assisted Gene Expression Restoration | *Machine Learning, Bioinformatics*

[GitHub](#)

- Explored nonlinear models for identifying the transcription factors that restore gene expression from a source cell state to a target state, extending the linear GAGER approach. Nonlinear models often reduced expression differences using fewer transcription factors across several scRNA-seq datasets.

Project Tokkhok | *React, FastAPI, PostgreSQL, Supabase, RAG*

[Demo](#) | [GitHub](#)

- AI-assisted learning platform for Bengali-speaking students: chat with an uploaded knowledge base (retrieval over a vector database) in Bangla, Banglish (Bangla typed in English letters), or audio, and a smart note editor that converts Banglish notes to proper Bangla.

Buldr-Bangladesh | *React, Spring Boot, PostgreSQL, Tailwind CSS, Firebase Auth*

[GitHub](#)

- Full-stack platform for engineers to learn (AI chatbot, smart notes, summarizing and paraphrasing tools), showcase projects, sell projects through a marketplace with order tracking, and get auto-generated portfolios.

Deepfake Image Detector Robust Against Adversarial Attacks | *PyTorch, ViT, YOLOv11*

[GitHub](#)

- Detector for real and AI-generated images (CIFAKE). Feature dropping and augmentation reached up to 80% accuracy on adversarial data, and custom adversarial noise and dropout layers improved robustness.

TechWorld | *Node.js, Express.js, PostgreSQL, Oracle PL/SQL*

[GitHub](#)

- E-commerce website for PCs with complex SQL and PL/SQL operations. Core feature: PC-BUILDER for assembling a custom PC from compatible parts.

Technical Skills

HCI Methods: Semi-structured interviews, reflexive thematic analysis, focus group discussions, formative studies

AI and LLMs: Agentic LLM pipelines, RAG, LLM-as-judge evaluation, web agents, PyTorch, fine-tuning transformer models

Languages: Python, C, C++, Java, JavaScript, Kotlin, SQL, Bash

Web and Databases: React, Tailwind CSS, Node.js, Express.js, Spring Boot, FastAPI, PostgreSQL, Supabase, Oracle DB

Tools: Playwright, Android Studio, Postman, Swagger, LaTeX

Achievements and Test Scores

- Champion, ITVerse 2023 "TechTales" Project Showcase Competition, University of Dhaka
- Finalist, BUET CSE Fest Capture the Flag (CTF) Competition 2023
- Dean's List Scholarship, BUET (Level 1 and Level 2)
- IELTS: 7.5 overall (Speaking 7.0)
- Competitive programming: 600+ problems solved, Codeforces max rating 1519 ([hay_re_manosh](#))

References

Available upon request.